THE DIMENSION OF EVERY COMPONENT OF AN INTERSECTION OF EQUIDIMENSIONAL CLOSED SUBSETS OF AFFINE SPACE IS AT LEAST R PLUS S MINUS N

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Dimension of a product of locally algebraic schemes and local dimension as the dimension of the local ring. Stacks Project, Varieties, Lemmas lemma-dimension-product-locally-algebraic (tag 0A2K) and lemma-dimension-locally-algebraic (tag 0A21). Supplies the local dimension of the product and the translation between local dimensions and component dimensions.
- Dimension drop by at most one for a quotient by a single equation. Stacks Project, Algebra, Lemma algebra-lemma-one-equation (tag 00OK). Gives the bound dim(R/(f)) >= dim(R) - 1 applied to the n linear forms.

Source reference. Original lemma 0B2M: sources/stacks-chow/tags.txt, source lines 13935–13935. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field and $n \geq 0$. Let $X, Y \subset \mathbf{A}^n_k$ be closed subsets. Assume that X and Y are equidimensional, $\dim(X) = r$ and $\dim(Y) = s$. Then every irreducible component of $X \cap Y$ has dimension $\geq r + s - n$.*

Proof. Consider the closed subscheme $X \times Y \subset \mathbf{A}^{2n}_k$ where we use coordinates $x_1, \ldots, x_n, y_1, \ldots, y_n$. Then $X \cap Y = X \times Y \cap V(x_1 - y_1, \ldots, x_n - y_n)$. Let $t \in X \cap Y \subset X \times Y$ be a closed point. By Lemma 0B2M we have $\dim_t(X \times Y) = \dim(X) + \dim(Y)$. Thus $\dim(\mathcal{O}_{X \times Y, t}) = r + s$ by Lemma 0A21. By Algebra, Lemma 00KW we conclude that

$$\dim(\mathcal{O}_{X \cap Y, t}) = \dim(\mathcal{O}_{X \times Y, t}/(x_1 - y_1, \ldots, x_n - y_n)) \geq r + s - n$$

This implies the result by Lemma 0A21. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.